

SimCalo: one row per particle that entered

`geantTrackId` of every PCaloHit = the particle at the calo surface (CaloSD)
⇒ **no clustering**: sum the hits per id

ECAL hits in GeV; HCAL hits relabelled and × digitiser sampling factors (GeV-equivalent); HF photoelectrons × HF factor (unvalidated)

cells summed over all particles first → optional per-cell threshold (rechit-like) → shared in proportion to the deposits

`response` = $E_{\text{deposit}}/E_{\text{at boundary}}$ (single particle)
`SimCaloJet` = anti-k_T 0.4 of the deposits (direction = energy-weighted cell direction)

SimPFCand: ideal particle flow

calo-entrance particle, charged, born inside the tracker volume, $|\eta| < 2.5$, $p_T > 0.5$ GeV
→ its **momentum** (perfect tracking), kind 1
otherwise → its **deposit** (neutral: kind 2, charged not trackable: kind 3), or nothing if it left none

`SimPFJet` = anti-k_T 0.4 of these; SimPF − SimCalo = what tracking recovers